Big O cheat sheet

Interview revision sheet

The topics that come up most in interviews, on one page. Typical costs, not guarantees: check the full sheet for assumptions.

Data structures

Topic	Time	Space
Dynamic array	Index O(1); append O(1) amortized; insert or delete in the middle O(n)	O(n)
Hash table (dict and set)	Lookup, insert, delete O(1) expected; O(n) worst case	O(n)
Stack	Push, pop, peek O(1)	O(n)
Queue and deque	Enqueue and dequeue O(1) with a deque	O(n)
Binary heap and priority queue	Peek O(1); push and pop O(log n); build from n items O(n)	O(n)
Binary search tree	Search, insert, delete O(h): O(log n) balanced, O(n) worst case	O(n)
Union-Find (disjoint-set union)	Amortized O(α(n)) per find or union	O(n)
Monotonic stack	O(n) total	O(n)

Sorting

Topic	Time	Space
Mergesort	Θ(n log n) in every case	O(n)

Searching

Topic	Time	Space
Binary search	O(log n)	O(1) iterative

Graphs and trees

Topic	Time	Space
Breadth-first search (BFS)	O(V + E)	O(V)
Depth-first search (DFS)	O(V + E)	O(V)
Tree traversals	O(n)	O(h) for depth-first orders
Dijkstra's algorithm	O((V + E) log V) with a binary heap	O(V + E)
Graph representations	Edge check: matrix O(1), list O(deg(u))	List O(V + E); matrix $O(V^2)$

Patterns

Topic	Time	Space
Two pointers	O(n)	O(1)
Sliding window	O(n)	O(1) or O(n)
Prefix sums	O(n) to build; O(1) per range sum	O(n)
Hash map frequency counting	Expected O(n) total	O(n)
Backtracking and subsets	$O(2^n)$ output generation	O(n) recursion, plus stored output
Dynamic programming	Number of states × transitions per state	Depends on memoization or tabulation
Merge intervals	O(n log n)	O(n)

Rough guide: input size to complexity

Input size n	Complexity that usually fits
Up to about 10	O(n!)
Up to about 20	$O(2^n)$
Up to about 500	$O(n^3)$
Up to about 5,000	$O(n^2)$
Up to about 1,000,000	O(n log n)
Up to about 100,000,000	O(n)

Assumes roughly 10^8 simple operations per second. Real limits depend on the language and the work per step.